MA3521 — Linear ODEs

Name: Answer Keys

Problem 1 $2y' + 2\ln(t)y = 2\exp(t - t\ln(t))$ $\quad y(1) = 0$

1a. Put the ODE in standard form $y' + p(t)y = q(t)$.

$p(t) =$ $\ln(t)$ $\quad q(t) =$ $e^{t - t\ln(t)}$

1b. Find an integrating factor.

$\mu(t) = \exp\left(\int \ln(t)\, dt\right) =$ $e^{t\ln(t) - t}$

1c. Multiply by $\mu(t)$ and integrate.

$(\mu y)' =$ 1

$\mu(t)y(t) = \int 1\, dt + C =$ $t + C$

1d. Use the initial condition to evaluate C.

$C =$ -1

$2y' + 2\ln(t)y = 2e^{(t - \ln(t))}$

$y' + \ln(t)y = e^{(t - \ln(t))}$

$p(t) = \ln(t)$

$q(t) = e^{(t - \ln(t))}$

1a

$M(t) = e^{\int \ln(t)\, dt}$

$\int \ln(t)\, dt = t\ln(t) - t$

$M(t) = e^{t\ln(t) - t}$

1b

$y' + \ln(t)y = e^{t - t\ln(t)}$

$e^{t\ln(t) - t}\, y' + e^{t\ln(t) - t} \ln(t)\, y = e^{t\ln(t) - t}\, e^{t - t\ln(t)}$

product rule

$= e^0$

$(My)' = 1$

$\int (My)'\, dt = \int 1\, dt$

$M(t)y(t) = t + C$

1c

$e^{t\ln(t) - t}\, y(t) = t + C$

$y(1) = 0$ gives $y = 0$, $t = 1$

$e^{1\cdot\ln(1) - 1} \cdot 0 = 1 + C$

$0 = 1 + C$

$-1 = C$

1d

Problem 2 $3y' + 3te^t y = 6t\exp\left((1-t)e^t\right) \qquad y(1) = 2$

2a. Put the ODE in standard form $y' + p(t)y = q(t)$.

$p(t) =$ te^t $\qquad q(t) =$ $2te^{(1-t)e^t}$

2b. Find an integrating factor.

$\mu(t) = \exp\left(\int te^t\, dt\right) =$ $e^{(t-1)e^t}$

2c. Multiply by $\mu(t)$ and integrate.

$(\mu y)' =$ $2t$

$\mu(t)y(t) = \int 2t\, dt + C =$ $t^2 + C$

2d. Use the initial condition to evaluate C.

$C =$ 1

$3y' + 3te^t y = 6te^{(1-t)e^t}$

$y' + te^t y = 2te^{(1-t)e^t}$

$p(t) = te^t$

$q(t) = 2te^{(1-t)e^t}$

1a

$M(t) = e^{\int te^t dt}$

$\int te^t dt \qquad u = t \quad dv = e^t dt$

$du = dt \quad v = e^t$

$= te^t - \int e^t dt$

$= te^t - e^t$

$= (t-1)e^t$

$M(t) = e^{(t-1)e^t}$

1b

$y' + te^t y = 2te^{(1-t)e^t}$

$e^{(t-1)e^t} y' + e^{(t-1)e^t} te^t y = e^{(t-1)e^t} 2te^{(1-t)e^t}$

product

$(\mu y)' = 2t$

$\int (\mu y)' dt = \int 2t\, dt$

$M(t)y(t) = t^2 + C$

1c

$e^{(t-1)e^t} y(t) = t^2 + C$

$y(1) = 2$ gives $y = 2, t = 1$

$e^{(1-1)e^1}(2) = 1^2 + C$

$e^0(2) = 1 + C$

$2 = 1 + C$

$1 = C$

1d